FOMULAE OF TRIGNO

1. $\sin x = \frac{1}{\cos ecx}$, $\cos x = \frac{1}{\sec x}$,$\tan x = \frac{1}{\cot x} = \frac{\sin x}{\cos x}$, $\cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}$
2. $\sin^2 x + \cos^2 x = 1$, $\sec^2 x - \tan^2 x = 1$, $\cos ec^2 x - \cot^2 x = 1$
3. $\sin(-x) = -\sin x$, $\cos(-x) = \cos x$, $\tan(-x) = -\tan x$
4. $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$, $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
5. $\cot(A \pm B) = \frac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$, $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$.
6. $2\sin A \cos B = \sin(A+B) + \sin(A-B)$, $2\cos A \sin B = \sin(A+B) - \sin(A-B)$.
7. $2\cos A \cos B = \cos(A+B) + \cos(A-B)$, $2\sin A \sin B = \cos(A-B) - \cos(A+B)$
8. $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$, $\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$.
9. $\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$, $\cos C - \cos D = 2\sin\frac{C+D}{2}\sin\frac{D-C}{2}$.
10. $\sin 2A = 2\sin A \cos A = \frac{2\tan x}{1+\tan^2 x}$, $\sin A = 2\sin\frac{A}{2}\cos\frac{A}{2} = \frac{2\tan\frac{A}{2}}{1+\tan^2\frac{A}{2}}$.
11. $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A = \frac{1-\tan^2 A}{1+\tan^2 A}$

    $\cos A = \cos^2\frac{A}{2} - \sin^2\frac{A}{2} = 2\cos^2\frac{A}{2} - 1 = 1 - 2\sin^2\frac{A}{2} = \frac{1-\tan^2\frac{A}{2}}{1+\tan^2\frac{A}{2}}$.
12. $\tan 2A = \frac{2\tan A}{1-\tan^2 A}$, $\tan A = \frac{2\tan\frac{A}{2}}{1-\tan^2\frac{A}{2}}$.
13. $1 + \cos 2A = 2\cos^2 A$. $1 + \cos A = 2\cos^2\frac{A}{2}$,$1 - \cos 2A = 2\sin^2 A$, $1 - \cos A = 2\sin^2\frac{A}{2}$.
14. $\sin 3A = 3\sin A - 4\sin^3 A$, $\cos 3A = 4\cos^3 A - 3\cos A$.
15. $\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$.
16. $\sqrt{1+\sin 2A} = \cos A + \sin A$, $\sqrt{1-\sin 2A} = \cos A - \sin A$
17. $\sqrt{1+\sin A} = \cos\frac{A}{2} + \sin\frac{A}{2}$, $\sqrt{1-\sin A} = \cos\frac{A}{2} - \sin\frac{A}{2}$.
18. $\frac{\cos A \pm \sin A}{\cos A \mp \sin A} = \frac{1 \pm \tan A}{1 \mp \tan A} = \tan\left(\frac{\pi}{4} \pm A\right)$, $\frac{\cos\frac{A}{2} \pm \sin\frac{A}{2}}{\cos\frac{A}{2} \mp \sin\frac{A}{2}} = \frac{1 \pm \tan\frac{A}{2}}{1 \mp \tan\frac{A}{2}} = \tan\left(\frac{\pi}{4} \pm \frac{A}{2}\right)$.
19. If $\sin x = 0$ then $x = n\pi$ and if $\sin x = \sin y$ then $x = n\pi + (-1)^n y$ $n \in Z$.
20. If $\cos x = 0$ then $x = (2n+1)\frac{\pi}{2}$ and if $\cos x = \cos y$ then $x = 2n\pi \pm y$, $n \in Z$

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21. If $tax = \tan y$ then $x = n\pi + y$, $n \in Z$.

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